Solar System Topological Lattice Gravitational Wave Observatory

(SS-TLGO)

Conceptual design for a very-very-wideband gravitational wave detector spanning the Solar System, using a topologically-protected quantum lattice as the measurement reference.

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Document version: 1.0
Document date: 2026-04-13
Status: Conceptual design, 50–100 year horizon. Engineering hand-waves are intentional and proportional to the timeframe.

This document describes a conceptual gravitational wave detector designed to be built when the supporting technology matures. The design is constrained by physics — strain sensitivity, frequency response, and quantum mechanical phase precision — but allows engineering capabilities (quantum networking, autonomous nanobot infrastructure, deep-space deployment) that do not currently exist but are on credible research trajectories. The document is structured to be a portable design summary that can be picked up by any sufficiently informed reader, including a future iteration of the project, a collaborator, or a re-engagement after a long pause.

1. Concept Summary

SS-TLGO is a *very-very-wideband* gravitational wave detector consisting of seven measurement stations distributed across the Solar System, connected by a topologically-protected quantum lattice that serves simultaneously as the phase reference, the entanglement distribution medium, and the quantum memory. The detector measures the integrated spacetime metric perturbation over its volume rather than length differences between fixed points.

The defining property of the detector is not its sensitivity at any single frequency — though that is also unprecedented — but its continuous spectral coverage. SS-TLGO is sensitive across approximately six orders of magnitude in gravitational wave frequency, from ~10 nHz at the lowest extreme to ~10 kHz at the highest, with no gaps. No existing or proposed instrument covers more than three orders of magnitude. The detector is the first GW instrument designed for source-agnostic discovery — instead of optimizing for known source classes (LIGO for stellar mergers, LISA for SMBH mergers, PTA for the stochastic background), SS-TLGO observes the entire spectrum and lets any source that emits in the band be detectable.

The instrument is built around three physical principles that distinguish it from all existing GW detectors:

- **Topological protection of the phase reference.** The quantum state encoding the QPLL phase lives in a topologically protected subspace of a distributed lattice, not in any individual qubit. Local decoherence events do not destroy the phase information; only correlated errors above the topological code's threshold do. This extends the effective coherence time of the measurement reference beyond what any physical qubit could sustain.
- **Closed-loop self-referencing across multiple nodes.** Each node's quantum phase-locked loop takes its phase reference from its neighbors via the entangled lattice. Common-mode drift in any single node cancels in the differential measurement across the closed loop. The detector has no central clock or absolute reference and is not limited by clock stability the way conventional GW detectors are.
- **Self-maintaining infrastructure via autonomous nanobot population.** The lattice's topological protection is sustained by a nanobot population that handles defect repair, environmental control, and component replacement faster than errors can accumulate. The nanobots are part of the detector, not an add-on. Without working autonomous infrastructure, the lattice cannot be maintained over the multi-decade integration times the science targets require.

These three principles, taken together, define a measurement instrument that is qualitatively different from any GW detector that has been built or proposed. The closest current analogs are atomic interferometer arrays (MAGIS-100, AION, AEDGE) which share the distributed-quantum-sensor concept but do not use topological protection or the closed-loop self-referencing architecture.

2. Node Placement

The detector consists of seven measurement stations distributed across the Solar System. The placement is chosen to maximize geometric leverage (long baselines for low-frequency sensitivity), dynamical stability (nodes at long-stable Lagrange or orbital points), and out-of-plane resolution (one node well out of the ecliptic for full 3D directional coverage).

Node	Location	Heliocentric distance	Purpose
N1	Sun-Earth L1	0.99 AU	Inner anchor; fast Earth comm; calibration reference
N2	Sun-Earth L2	1.01 AU	Inner anchor; pairs with N1 for short-baseline cross-check
N3	Sun-Jupiter L4	5.2 AU	Long-stable Trojan; intermediate baseline
N4	Sun-Saturn L4	9.5 AU	Long-stable Trojan; mid-outer baseline
N5	Sun-Neptune L4	30.1 AU	Long-stable Trojan; outermost ecliptic baseline
N6	Sun-Neptune L5	30.1 AU	Outer pair with N5 for redundant outer triangulation
N7	Pluto orbit	39.5 AU mean	Out-of-ecliptic node; provides z-vector resolution

2.1 Geometric rationale

N1 and N2 at the Sun-Earth Lagrange points provide a sub-AU baseline (~3 million km) for the highest frequency band and serve as the calibration anchor. They are reachable by conventional spacecraft for maintenance and direct correlation with Earth-based ground stations. *These nodes are also the testbed phase* — the L1/L2 pair will be deployed first and operated as a small-scale demonstration of the architecture before commitment to the outer-planet infrastructure. The full SS-TLGO is built outward from this validated core.

N3, N4, N5 sit at the most dynamically stable Trojan points of Jupiter, Saturn, and Neptune. Trojan L4/L5 points are gravitationally stable on Gyr timescales — the real Trojan asteroid populations have been there since the early Solar System — so the nodes do not need active station-keeping in the orbital sense, only in the local geometric sense for lattice maintenance.

N6 at Neptune L5 pairs with N5 to give two outer Trojan nodes, providing redundant long baselines at the outermost ecliptic radius and protecting against the loss of either single outer node.

N7 in orbit around Pluto is the critical out-of-plane node. Pluto's orbital inclination of ~17° to the ecliptic gives it significant z-displacement at most points in its orbit. Without N7, the detector is effectively a flat hexagonal array in the ecliptic plane with directional nulls perpendicular to the plane. N7 lifts the geometry into 3D and removes the nulls, providing full 4π steradian sky coverage.

2.2 Baseline inventory

Six discrete (or semi-discrete) ecliptic-plane nodes plus one out-of-plane node yield 21 unique baselines ($n \times (n-1) / 2$ for $n = 7$). The most important for understanding the frequency response are listed below.

Baseline	Length	Optimal sensitivity band	Notes
N1 ↔ N2	0.02 AU (3×10^6 km)	mHz – kHz	Highest frequency, testbed pair
N1 ↔ N3	~5 AU	µHz – mHz	Inner-to-Jupiter baseline
N3 ↔ N4	~7 AU	µHz – mHz	Jupiter-to-Saturn baseline
N4 ↔ N5	~22 AU	sub-µHz	Saturn-to-Neptune baseline
N3 ↔ N5	~28 AU	sub-µHz	Cross-Sun outer baseline
N5 ↔ N6	~30 AU chord	sub-µHz	Neptune Trojan pair
N1 ↔ N7	~38 AU + z	sub-µHz, 3D	Inner-to-Pluto, out of plane
N5 ↔ N7	varies, ~10–30 AU	sub-µHz, 3D	Outer-to-Pluto, full 3D

The frequency response of each baseline peaks where the GW wavelength matches roughly the baseline length. The collection of baselines spans approximately six orders of magnitude in frequency, from ~10 nHz at the longest cross-Sun baselines up to ~10 kHz at the testbed N1↔N2 pair.

3. Quantum Lattice Architecture

Each node hosts a section of a topologically-protected qubit array. The local lattice at each node is a 3D arrangement of physical qubits maintained at cryogenic temperatures inside a passively-shielded enclosure. The full detector lattice is the union of all node-local lattices plus the inter-node entanglement links carried by the quantum repeater chains.

3.1 Physical qubit realization

The candidate physical implementations all involve solid-state defect or trapped-particle systems with long ground-state coherence and good optical coupling. The leading candidates are:

- Diamond NV centers — well-developed technology, room-temperature coherence demonstrated, optical interface mature.
- Silicon vacancy centers — better optical coherence than NV, requires cryogenic operation, recently demonstrated in repeater applications.
- Rare-earth-doped crystals (Er, Eu, Pr in Y_2SiO_5) — best demonstrated quantum memory coherence times (hours at cryogenic temperatures with magnetic field decoupling), optical telecom-band compatibility.
- Trapped-ion arrays — best coherence and gate fidelity but harder to scale to lattice geometries.
- Neutral atoms in optical tweezer arrays — recent rapid progress, scalable, good coherence.

The choice between these is a 25-year-out engineering decision. The design assumes a system in the rare-earth or silicon-vacancy class, achieving >1000-second physical coherence at high fidelity, with topological protection extending the logical coherence beyond that.

3.2 Topological protection

The qubit array runs a surface code or color code at sufficient distance to suppress local errors below the threshold where the nanobot infrastructure can repair them faster than they accumulate. The logical qubits encoding the QPLL phase reference live in the topologically protected subspace, not in any physical qubit.

This is the architectural move that makes the coherence problem tractable. Instead of trying to keep individual qubits coherent for hours or days, the phase information is stored in a collective topological degree of freedom that is intrinsically protected against the dominant decoherence channels. A single-qubit decoherence event does not destroy the phase information — only correlated errors above the code's correction threshold do, and those are exactly what the active error correction (driven by the nanobot infrastructure) is designed to suppress.

The choice of topological code (surface code, color code, Fibonacci anyon model, or others) is a design parameter that trades off code distance against physical qubit overhead and correction speed. The design assumes a surface code at distance ~50, requiring ~10^4 physical qubits per logical qubit. With ~10^9 physical qubits per node (an aggressive but plausible target for 2050 fabrication), each node hosts ~10^5 logical qubits available for the QPLL operation and inter-node entanglement.

3.3 Inter-node entanglement distribution

Entanglement between nodes is established and maintained through a chain of quantum repeater nodes distributed along each baseline. Photonic entanglement is generated locally at each station, transmitted via narrow optical beams to neighboring stations or repeater relays, and stored in the local quantum memory upon arrival.

The repeater chain spacing is set by photon loss budget. At interplanetary distances with 10-meter aperture optics in clean line-of-sight conditions, geometric loss alone is ~$1/d^2$, and over 1 AU from a 1-meter emitter to a 10-meter receiver this is approximately 10^{-22} — far too lossy for direct transmission. Repeater spacing of approximately 10^7 to 10^8 km, with ~10–100 repeater nodes per long baseline, brings the loss budget into a workable range.

The repeater nodes are deployed and maintained by autonomous nanobot infrastructure. Each repeater is a small autonomous spacecraft with its own quantum memory, optical apertures, and entanglement-generation hardware. The repeater chain is the most physically distributed component of the detector and the one most exposed to deep-space hazards (cosmic rays, micrometeoroids, thermal cycling).

3.4 The role of the nanobots

The nanobot infrastructure is part of the detector, not an add-on. Topological protection works only as long as the underlying physical lattice maintains its connectivity and the error rate stays below the correction threshold. In a hostile environment (interplanetary space, with solar wind, cosmic rays, thermal cycling, micrometeoroids), the lattice continuously accumulates damage, and that damage must be repaired faster than it accumulates.

A static lattice fails. A self-maintaining lattice — repaired by autonomous nanoscale machinery that operates as part of the system — works. The nanobot population is the immune system of the lattice. Their tasks include:

- Continuous identification and repair of physical qubit errors before they correlate into logical errors
- Replacement of failed components in the local qubit arrays
- Maintenance and replenishment of cryogenic cooling fluids and shielding materials
- Repair and recalibration of optical components and quantum repeater nodes along the baselines
- Manufacturing of replacement components from local materials (in-situ resource utilization)
- Self-replication of the nanobot population itself, with error-correction to prevent runaway evolution or drift

The nanobot population is the single biggest engineering uncertainty in the design. See Section 8 for the full assessment.

4. The Quantum Phase-Locked Loop

The QPLL is the measurement instrument. Each node runs a quantum phase-locked loop whose reference is the topological phase shared with its neighboring nodes via the entangled lattice connection. Each node has at minimum two neighbors (some have more, depending on which baselines are active at any given moment), enabling closed-loop self-referencing that removes common-mode drift from the differential measurement.

4.1 Clock rate set for stability, not maximum speed

The QPLL clock rate is set for stability and well above the local atomic clock rate, but only as fast as needed to oversample the highest GW frequency of scientific interest with comfortable margin. Oversampling beyond that range is computational waste with no scientific return.

For the SS-TLGO target of ~10 nHz to ~10 kHz coverage, a QPLL clock rate of approximately 10 kHz to 100 kHz is sufficient. This is well above the typical optical atomic clock comparison rate of ~10 Hz, providing comfortable headroom for atomic-clock cross-checking of the QPLL phase, while staying below the rates that would generate impractical data volumes.

The local atomic clock at each node serves as an independent diagnostic of the QPLL coherence. If the QPLL phase drifts relative to the atomic clock at rates inconsistent with the expected GW signal frequencies, the topological protection is degrading and the nanobots intervene. This is a continuous hardware self-test built into the operating principle, requiring no external command, and is essential for autonomous multi-decade operation.

4.2 Differential measurement

The measured quantity at each baseline is the time-varying differential phase between the two endpoint nodes, integrated over the GW period of interest. A passing gravitational wave shows up as a coherent perturbation across multiple baselines with the directional and polarization signature of the wave.

With three or more nodes in a closed loop, the closed-loop phase relationships across the topology cancel any drift that is common to all baselines (e.g., drift in any single node's local oscillator, common environmental perturbations of the lattice). What survives is the part of the phase variation that is genuinely differential between baselines — exactly what a GW produces, and almost nothing else.

4.3 Massive oversampling and noise integration

At the chosen QPLL clock rates of 10–100 kHz, each cycle of a 1 µHz GW signal is sampled approximately 10^{10} to 10^{11} times. Even at the lowest GW frequencies of interest (~10 nHz), each cycle is sampled ~10^{13} to 10^{14} times over its 3-year period. The phase noise floor can be characterized in extraordinary detail and signals at the noise floor extracted by integration over the multi-decade measurement cycle.

This is the same principle that makes pulsar timing work — slow signal, fast samples, integrate forever — but applied to a coherent quantum phase measurement instead of a classical pulse-arrival-time measurement.

5. Data Architecture

With the QPLL clock rate set for stability rather than maximum speed (Section 4.1), the data volumes drop substantially from earlier estimates. This is a meaningful simplification of the design.

5.1 Per-node raw data rates

Each node generates raw phase data at the QPLL clock rate × number of active baselines × phase precision metadata. For a node with ~10 active baselines and a clock rate of ~100 kHz, this is approximately 1 megasamples/second × 10 baselines × ~16 bytes per sample ≈ ~160 MB/s ≈ ~1.3 Gbits/s per node. With seven nodes, total raw data is approximately 10 Gbits/s across the network. This is well within the envelope of plausible deep-space optical communications by the design horizon.

Higher-rate operation (kHz GW band, MHz QPLL clock) generates correspondingly more data. The detector can be operated in modes that prioritize high-frequency or low-frequency science depending on which baselines are sampled at which rates. The total data envelope is bounded by available downlink rather than by lattice physics.

5.2 Onboard processing

Each node runs a real-time correlator that combines its local phase data with arrivals from its neighbors via classical optical channels (with light-time delays applied), producing reduced data products at much lower rates — typically megabits per second of processed phase residuals per baseline. The reduction strategy is straightforward: most of the raw data is noise that can be discarded once the noise floor is characterized, and the science signal is in the residuals after subtracting modeled noise sources.

The required compute capability per node is comparable to current radio interferometry correlators (e.g., ALMA, SKA), well within the trajectory of space-rated computing over 25–50 years.

5.3 Earth-network downlink and ground correlator

Reduced data products from each node are downlinked to Earth-based facilities at megabit-per-second rates per node, totaling ~100 Mbits/s across the full network. This is well within the envelope of current and projected deep-space optical communications (NASA's Deep Space Optical Communications experiment in 2023 demonstrated gigabit-class downlink from Mars distances).

The ground correlator combines data from all seven nodes coherently, producing the final GW phase residuals across all 21 baselines. With 21 baselines and sample rates to ~100 kHz, the correlator workload is comparable to current radio interferometry correlators extended to many more baselines and far longer integration times.

5.4 Long-term data preservation

The detector's planned measurement cycle is 60 years minimum, 180 years ideal. Coherent integration over these timescales requires preserving calibrated phase data for the full duration. This is a fundamentally new data regime — no current scientific instrument operates with calibration models intended for multi-century coherent integration.

The data preservation challenge is real but is in the domain of planetary-scale long-term archival, which is being worked on for other reasons (climate records, genomic databases, cultural heritage). The detector benefits from those parallel efforts and does not need a bespoke solution.

6. Operating Envelope Summary

The detector's nominal operating parameters, quoting the realistic mid-case where the answer depends on engineering details:

Parameter	Value
Frequency coverage	~10 nHz to ~10 kHz (continuous, no gaps)
Optimal sensitivity band	~μHz to ~mHz
Strain sensitivity (optimistic floor)	$\sim 4 \times 10^{-31}$
Strain sensitivity (realistic)	$\sim 10^{-26}$ to $\sim 10^{-28}$
Sky coverage	Full 4π steradians, no nulls
Source localization (bright sources)	~degree precision
Source localization (continuous, integrated)	~arcminute
Polarization decomposition	Full GR tensor (h+, h×) plus alternative-gravity polarizations
Measurement cycle (minimum)	60 years
Measurement cycle (ideal)	180 years
Number of nodes	7
Number of baselines	21
QPLL clock rate	~10 kHz to ~100 kHz (set for stability)
Per-node raw data rate	~1 Gbit/s
Network reduced data rate to Earth	~100 Mbits/s

7. Capability Metric

This section quantifies what the detector can actually measure and detect. Numbers are order-of-magnitude estimates appropriate to the engineering uncertainty of a 50–100 year design, with optimistic and conservative envelopes given side by side where the answer depends on subsystem assumptions.

7.1 Strain sensitivity calculation

The strain sensitivity h(f) of the detector at GW frequency f is set by the phase precision δϕ achievable by the QPLL, the baseline length L, and the integration time T. The basic relation:

$$h(f) \approx (\delta\phi \times \lambda) / (2\pi \times L \times \sqrt{N\_samples})$$

where λ is the optical wavelength of the QPLL light (~1 µm), L is the baseline length, and N_samples is the number of independent phase samples integrated.

For a topologically-protected QPLL operating at the Heisenberg limit with realistic photon numbers (N_photon ~ 10^{12} per second per link, achievable with current-generation entangled photon sources scaled to 25-year technology), the per-sample phase precision is approximately:

$$\delta\phi\_single \approx 1/\sqrt{N\_photon} \approx 10^{-6} \text{ radians per second of integration}$$

Over a 60-year measurement cycle the integrated phase precision approaches:

$$\delta\phi\_integrated \approx 10^{-6} / \sqrt{(60 \times 3 \times 10^{7})} \approx 2 \times 10^{-11} \text{ radians}$$

Combining with the longest baseline (~60 AU between N3 and N5 across the Sun) and the optical wavelength:

$$h\_min(f\_optimal) \approx (2 \times 10^{-11} \times 10^{-6}) / (2\pi \times 9 \times 10^{12} \text{ m}) \approx 4 \times 10^{-31}$$

This is the optimistic floor — the best possible strain sensitivity at the optimal frequency over a 60-year integration time. For an ideal 180-year measurement cycle, the sensitivity floor drops by a further factor of √3 ≈ 1.7.

7.2 Comparison to existing and planned detectors

Detector	Optimal frequency	Strain sensitivity at peak
LIGO (current, 2026)	~100 Hz	~10^{-23}
Einstein Telescope (planned)	~10 Hz	~10^{-24}
LISA (planned, ~2035)	~1 mHz	~10^{-20}
NANOGrav 15-yr (current)	~1 nHz	~10^{-14} (effective)
SKA-PTA (planned)	~1 nHz	~10^{-16} (effective)
SS-TLGO floor (this design)	~µHz to mHz	~4×10^{-31}

Detector	Optimal frequency	Strain sensitivity at peak
SS-TLGO realistic (this design)	~μHz to mHz	~10^{-26} to 10^{-28}

Even the conservative realistic estimate makes SS-TLGO 5–7 orders of magnitude more sensitive than LISA in the same frequency range, and 10–13 orders of magnitude more sensitive than current PTAs in the nanohertz band. The optimistic floor is, on paper, the most sensitive measurement of any physical quantity ever proposed.

7.3 Frequency coverage breakdown

Frequency	GW wavelength	Best baseline	Science target
~30 nHz	~30 light-years	N3⟷N5 cross-Sun (60 AU)	Stochastic background, longest SMBH inspirals
~1 μHz	~1 light-year	Multiple long baselines	Optimal band, individual SMBH binaries
~1 mHz	~10^9 km	N1⟷N3 (5 AU)	LISA overlap, EMRI sources
~1 Hz	~3×10^5 km	N1⟷N2 (3×10^6 km)	Compact binaries, LIGO low-frequency overlap
~10 kHz	~30 km	Local QPLL coherence	Hardware-limited upper edge

The detector is sensitive across approximately six orders of magnitude in GW frequency, with optimal strain sensitivity in the sub-μHz to mHz band where the long baselines and QPLL coherence both contribute maximally. Above ~1 Hz the sensitivity degrades because the GW wavelength becomes shorter than the baseline. Below ~10 nHz the frequency is below what 60 years of integration can resolve in one cycle.

7.4 Observable populations and science targets

With strain sensitivity in the 10^{-26} to 10^{-28} range over the optimal band, the observable source populations are dramatically expanded relative to current and proposed detectors. The targets below are listed in approximate order of scientific impact.

Supermassive black hole binaries

- All SMBH binaries in the observable universe with masses above ~10^5 M☉, throughout their entire inspiral history (not just the last few years before merger). Where current PTAs see only the brightest sources during the last ~thousand years before merger, SS-TLGO sees individual sources for millions of years.
- Continuous tracking of inspiral evolution for individual sources, allowing direct measurement of mass, spin, eccentricity, and orbital evolution from the early inspiral phase.

- Cosmological evolution of the SMBH binary population as a function of redshift, sampling the full merger history of structure formation rather than just the present-day rate.

Stellar-mass compact binaries

- All stellar-mass compact binaries in the Milky Way, throughout their entire inspiral history from formation to merger. The current LIGO catalog sees only the last ~seconds of life for binaries that happen to merge during the observation window. SS-TLGO sees the full population continuously.
- Galactic-scale census of compact binary evolution, providing direct constraints on stellar evolution, common envelope physics, and the formation pathways of compact objects.

Cosmological gravitational wave backgrounds

- Direct detection (not just upper-limit constraints) of the inflationary GW background, if it exists at any plausible amplitude.
- Direct detection of GW backgrounds from cosmological phase transitions (electroweak, QCD, GUT-scale), with enough frequency resolution to distinguish their spectra.
- Direct detection of GW backgrounds from cosmic strings or other topological defects.
- Possible detection of GWs from the cosmological neutrino background's clustering on small scales. Speculative but the sensitivity envelope reaches it.

Tests of general relativity

- Sensitivity to alternative GW polarizations (scalar breathing, vector longitudinal) at amplitudes that would distinguish GR from essentially all proposed alternative theories.
- Frequency-dependent dispersion measurements of GWs, testing Lorentz invariance and graviton mass at unprecedented precision.
- GW echoes from black hole horizons, testing quantum-corrections-to-GR scenarios at amplitudes 5+ orders of magnitude below current bounds.

Speculative but accessible targets

- Quantum gravity signatures in the GW background, if any exist with amplitudes above ~10^{-28} in the optimal band. Whether such signatures exist is unknown, but SS-TLGO is the first instrument that could look.
- Dark matter signatures in the form of GW emission from compact dark matter objects (primordial black holes, boson stars, dark stars), including direct detection of dark matter passing through the Solar System if it is in compact form.
- Direct detection of exotic compact objects (gravastars, fuzzballs, traversable wormholes) at amplitudes that would distinguish them from black holes.
- The original SGWB / nanohertz clock-error question that motivated this entire line of work. At SS-TLGO sensitivity, any nanohertz-band signal of plausible cosmological or local origin would be detectable.

7.5 The bottom line

SS-TLGO would observe gravitational waves from essentially every source that emits them in the observable universe, at sufficient sensitivity to characterize the source physics in detail rather than just detect the signal. The instrument transforms gravitational wave astronomy from a discovery science (finding new sources) to a precision science (measuring known sources to understand their physics). The closest analogy is the transition in optical astronomy from photographic plates to space-based

precision astrometry — same wavelength, same sky, completely different scientific posture and an order of magnitude or more of new discovery space.

The detector also opens the entire μHz–mHz frequency band, currently inaccessible to any planned instrument. This is the band where the universe's most massive bound systems live: galactic-scale gravitational dynamics, the largest SMBH binaries, the cosmological GW background's main features, and any new physics that operates at long timescales and large length scales. The fact that this band is currently dark is the biggest single observational gap in gravitational wave astronomy, and SS-TLGO's principal scientific justification is filling it.

8. Engineering Challenges

The engineering challenges below are organized by physical subsystem, with an honest assessment of where each one sits on a tractable-vs-open spectrum. The categorization uses three levels:

- **TRACTABLE:** an extension of current engineering trajectories, plausibly solvable with the resources of a major program over the design timeframe
- **HARD:** requires significant new physics or engineering breakthroughs, but the path is at least visible from current trajectories
- **OPEN:** no current path; would require fundamental advances we cannot currently anticipate

Note on scope: this section addresses physical and technical challenges only. Funding, political, and institutional questions are not engineering problems and are excluded by design. The detector either works as a piece of physics or it does not; whether it gets built is a separate question that the design document does not need to answer.

8.1 Quantum hardware

- **Topological lattice qubits at scale** — HARD. Current topological qubit experiments work with single logical qubits or small arrays. The detector requires lattices of millions to billions of physical qubits per node, encoding thousands of logical qubits running surface or color codes at high distance.
- **Quantum memory coherence at hour-to-day scale** — HARD. Current memories reach minutes in the best lab conditions. Hour-scale coherence is a known target of the quantum repeater community for the global quantum internet, and is a credible 20-year goal.
- **Quantum repeater chains at AU baselines** — HARD. Current quantum repeaters work at km to ~100 km distances. Scaling to ~10^7–10^8 km between repeaters in interplanetary space, with tens of repeater nodes per long baseline, requires both the coherence advances above and a reliable manufacturing-and-deployment infrastructure.
- **Photon source rate and fidelity** — TRACTABLE. Single-photon and entangled-photon sources at MHz rates with >99% fidelity already exist in lab settings.

8.2 Cryogenics and thermal management

- **Sustained cryogenic operation at outer planet distances** — HARD shading toward TRACTABLE. NV centers and similar solid-state qubits typically need temperatures below 4 K. At Neptune's distance the natural radiative equilibrium temperature is ~50 K, which gets you part of the way for free, but final cooling stages require active refrigeration. Current spacecraft cryocoolers operate at ~6 K for ~10 year lifetimes; the detector needs lower temperatures and longer lifetimes.
- **Thermal isolation from solar input** — TRACTABLE at outer nodes (low solar flux), HARD at inner nodes (N1, N2 see ~1 kW/m² that must be excluded from the cryogenic enclosure).
- **Local thermal stability of the lattice** — HARD. Topological protection works only if the thermal environment is below the topological gap. Maintaining temperature stability at the millikelvin level over decades, with only autonomous nanobot intervention for repairs, is beyond current spacecraft thermal control by several orders of magnitude.

8.3 Power supply

- **Multi-decade power generation at outer planet distances** — HARD. Solar power scales as $1/d^2$ and is essentially useless beyond Jupiter. The outer nodes must use radioisotope power systems, compact fission reactors, or beamed power.
- **Pu-238 inventory** — HARD. For a node needing ~10 kW continuously, ~500 kg of Pu-238 per node is required, with inventory replacement on ~50-year cycles to compensate for decay. Current global Pu-238 production is ~1.5 kg/year worldwide, requiring a dedicated multi-decade production program for the SS-TLGO inventory.
- **Compact fission reactors in space** — HARD shading toward TRACTABLE. NASA's Kilopower program and historical Russian reactors provide ~kW class power with ~10-year design lifetimes. Scaling to 10 kW class for 100+ year operation requires fuel reprocessing or fresh fuel delivery, plausible with nanobot infrastructure but not currently demonstrated.
- **Realistic answer** — a combination: compact fission reactors as primary power, RTGs as backup and for low-power components, nanobot-mediated refueling and component replacement on roughly 50-year cycles.

8.4 Onboard processing

- **Real-time correlation and data reduction** — TRACTABLE. Each node generates ~1 Gbit/s raw and reduces to ~megabits/s of useful science data through onboard correlation. Compute requirements are comparable to current radio interferometry correlators, well within the trajectory of space-rated computing over 25–50 years.
- **Radiation tolerance over multi-decade operation** — HARD. Current rad-hard processors are designed for ~10–20 year missions. Self-repairing electronics with nanobot-mediated component replacement, biological computing, or fully redundant systems are the candidates.
- **Algorithmic continuity over 60–180 years** — HARD. The data processing pipelines, calibration models, and analysis software for a 180-year mission must remain interoperable across many generations of computing technology. The detector needs either an unprecedentedly stable computing standard or a continuous translation layer maintained over the mission lifetime.

8.5 Communications

- **Inter-node classical communication** — TRACTABLE. Optical communication at gigabit rates over interplanetary distances exists in current technology demonstrations (NASA DSOC 2023). Scaling to higher rates is within reach.
- **Quantum entanglement distribution** — see Section 8.1.
- **Earth-network downlink for science data** — TRACTABLE at the ~100 Mbits/s reduced data product rate.

8.6 Station-keeping and lattice stability

- **Trojan point station-keeping** — TRACTABLE. Lagrange L4/L5 points are dynamically stable on Gyr timescales for small perturbations, requiring minimal active station-keeping. L1/L2 points are unstable on month timescales but standard halo orbit techniques handle this with low fuel budgets.
- **Pluto orbit maintenance** — TRACTABLE. Low-thrust ion propulsion with nanobot-mediated propellant replenishment.

- **Lattice geometric stability at the wavelength scale** — HARD. The QPLL phase reference depends on knowing inter-node distances to a fraction of the optical wavelength (~100 nm). Real-time geometric calibration via laser ranging between nodes, combined with high-precision orbital determination, is the standard approach. The challenge is maintaining sub-wavelength geometric knowledge over 60–180 years.

8.7 Nanobot infrastructure

- **Autonomous self-maintenance over centuries** — OPEN. Current nanotechnology produces functional nanostructures (DNA origami, molecular machines) but the gap between current capabilities and 'autonomous self-replicating maintenance robots operating in deep space for 200 years' is large. The 25-year horizon is optimistic; 50 years is more credible. **This is the biggest single engineering uncertainty in the design.**
- **Manufacturing at scale at the deployment site** — OPEN. The nanobot population needs to be either manufactured before launch (mass-prohibitive) or manufactured in situ from local materials (asteroids, comet ices, captured solar wind). In situ resource utilization at the nanoscale does not currently exist as engineering practice, though it is being worked on as part of the broader off-Earth presence (Mars colonization, asteroid mining) which the SS-TLGO design assumes will exist by the deployment timeframe.
- **Nanobot reliability and reproduction** — OPEN. Self-replication of nanoscale machinery has been demonstrated only in trivial cases. Reliable, error-correcting, evolution-resistant self-replication at infrastructure scales is a long-term research goal.

8.8 Operational security and integrity

The detector is a multi-decade autonomous system with quantum communication links, nanobot infrastructure with self-modification capability, and processing nodes at multiple Solar System locations. Each of these is an attack surface.

The lattice's topological protection makes some attacks harder — local decoherence cannot trivially destroy the logical state — but other attacks are entirely new. Corrupted nanobots can self-replicate. Quantum communication links may be subject to side-channel attacks that classical comms are not vulnerable to. An autonomous system with a 180-year lifetime that no human directly controls is the textbook example of 'set it and forget it' becoming 'set it and lose it' if the security architecture is not right from the start.

The architectural lesson from past hardware vulnerabilities (Intel ME, Stuxnet's PLC firmware exploitation, similar) is that any system that includes a privileged layer the main system cannot inspect or control creates a permanent vulnerability, regardless of how good the upper layers are. For SS-TLGO this means the nanobot self-replication and self-modification capability must be inspectable and constrained at the design level, not delegated to runtime trust. The nanobot population's behavior must be defined by hardware-rooted constraints that cannot be overridden by software, even by software running on the nanobots themselves.

Categorization: HARD, in the same class as the physics and engineering challenges, requiring upfront design work integrated with the rest of the architecture.

8.9 Summary table

Subsystem	Hardest single challenge	Category
Quantum hardware	Topological qubit lattice at scale	HARD
Coherence	Hour-to-day quantum memory at high fidelity	HARD
Cryogenics	Sub-K stability for centuries with autonomous repair	HARD
Power	Multi-decade kW-class power at Neptune	HARD
Processing	Algorithmic continuity over 60–180 years	HARD
Communications	All sub-systems within tractable envelope	TRACTABLE
Station-keeping	Sub-wavelength geometric stability over decades	HARD
Nanobots	Autonomous self-replicating infrastructure	OPEN
Security	Hardware-rooted constraints on self-modifying infrastructure	HARD

Only one challenge is genuinely OPEN: autonomous nanobot infrastructure. Everything else is HARD but at least visible from current trajectories. The detector cannot be built without solving the OPEN challenge, and that challenge depends on fundamental advances in nanotechnology that are being pursued in parallel for many other reasons (medicine, materials manufacturing, the broader off-Earth presence).

The 25-year horizon is optimistic for solving the open challenge in isolation. A 50–100 year horizon is more credible, and is consistent with the broader expectation that humanity's expansion into the Solar System will proceed at a pace that makes deployment of an SS-TLGO-class instrument feasible by the time the supporting technology matures. The detector is part of a larger trajectory, not an isolated mission.

9. Closing Notes

9.1 What the design is, and what it is not

This document describes a conceptual gravitational wave detector at the level of detail appropriate to a 50–100 year horizon. It is not a buildable specification. The engineering hand-waves are intentional and proportional to the timeframe. The purpose of the document is to establish that the detector is physically possible, internally consistent, and motivated by science targets that no current or planned instrument can address.

The physics is sound: the strain sensitivity is bounded by quantum mechanics (the Heisenberg limit on phase measurement) and the baseline length, both of which are calculable from first principles. The realistic operating sensitivity is limited by engineering, and is what the design effort is for. There is no fundamental obstacle to building a detector with these capabilities, only engineering challenges that follow current research trajectories at a pace that is plausible over 50–100 years.

9.2 Why this design and not LISA-bigger or PTA-bigger

SS-TLGO is qualitatively different from existing or planned detectors in three ways:

- **Topological protection of the phase reference** decouples the measurement from any single qubit's coherence, allowing the multi-decade integration that the lowest-frequency science targets require. No conventional interferometer or PTA has this property.
- **Closed-loop self-referencing** removes the need for a stable absolute clock or reference. The detector is not limited by clock stability the way conventional detectors are, and the thorny barycentric and clock-comparison issues that affect PTAs do not apply.
- **Six orders of magnitude continuous frequency coverage in one instrument** is the principal scientific value. No conventional approach scales this way — interferometers and PTAs each cover one band, and combining them does not give continuous coverage because of the gaps between bands.

These three properties are not incremental improvements over existing designs. They define a different category of instrument.

9.3 The relationship to the original SGWB question

This design originated as a response to the question that started a much earlier line of investigation: whether nanohertz-band signals from poorly understood sources, including possible clock-error or unmodeled-physics signatures, exist in the GW spectrum and can be detected. The pre-registered planetary harmonic analysis (documented separately) was a constrained attempt to look for such signals using existing data and produced rigorous null results in the regimes accessible to existing instruments.

SS-TLGO is the instrument that would answer the original question. At the design sensitivity, any nanohertz-band signal of plausible cosmological or local origin would be detectable, including signals that current PTAs cannot resolve due to baseline limitations and signals that LISA cannot reach due to its frequency band. The design is therefore not a separate project but the natural extension of the original line of work onto the timescale and infrastructure required to actually answer the question.

9.4 Status

Conceptual design, drafted as a thought experiment in the context of a long collaborative discussion. Not submitted, not peer-reviewed, not endorsed by any institution. The author retains all rights and may develop, abandon, or share the design as he sees fit. The document exists primarily to preserve the architecture and reasoning in a form that can be returned to or shared without losing context.

Comments, criticisms, refinements, or alternative designs are welcome. The author can be reached at the email address on the cover page.

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